Online News Popularity Analysis

Predicting Social Media Engagement using Machine Learning and Data Mining Techniques on the UCI Mashable Dataset.

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DATASET OVERVIEW

39,797 (39644)

Total Articles

61 (58 + 2 +1)

Feature Count

Key Feature Categories

- **Content Stats:** Word counts and article length
- **Popularity:** Keyword popularity metrics
- **Sentiment:** Subjectivity and polarity scores
- **Target Variable:** Number of Shares
- **Timing:** Publication day and weekend flags

MULTI-METHOD APPROACH

Supervised

Classification and Regression to predict popularity and share counts.

Unsupervised

Clustering articles using K-Means and DBSCAN to find natural groups.

Data Mining

Association Rule Mining (Apriori) for feature relationship discovery.

Analytics

Dimensionality reduction and Big Data processing via PySpark.

Big Data Analytics

Classification
- LR, RF

Regression
- Linear, Ridge, Lasso

Clustering
- K means, DBScan

Association Rule Mining
- Apriori

Dimensionality Reduction
- PCA, t-SNE

Temporal Analysis
- Average, Median

Feature Importance Analysis
- Gini Importance, Permutation

Exploration + Prediction

Analytics from ML

Classification (8 models)

- LR, KNN, SVM, RF, Decision Tree, Gradient boosting, Adaboost, Naive Bayes

Regression (7 models)

- Linear, Ridge, Lasso, Decision Tree Regressor, Gradient Boosting Regressor, SVR

PySpark Pipeline

Scalable Big Data Implementation

Built using **PySpark for distributed processing**

- Pipeline: Data → Features → Scaling → Random Forest
- Enables **parallel computation on large datasets**

Ensures Scalable + consistent results

EXPLORATORY DATA ANALYSIS

Key Findings

- **Skewed Distribution:** Share counts are highly positively skewed.
- **Elite Content:** Only a few articles achieve viral status.
- **Length vs shares:** Weak correlation between word count and engagement.
- **Temporal Patterns:** Weekend articles consistently see higher average shares.

Figure 3: Correlation Heatmap of Numeric Features

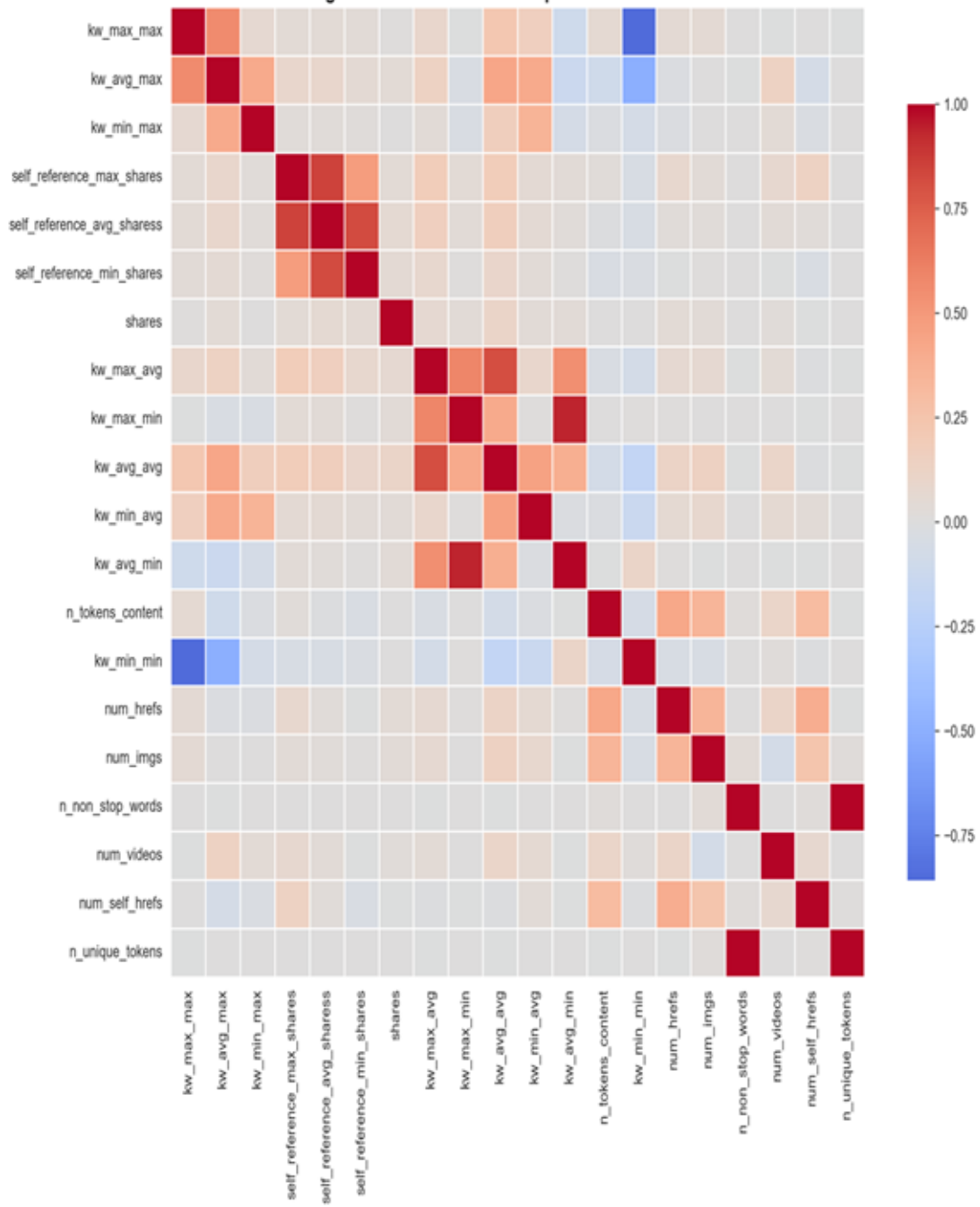
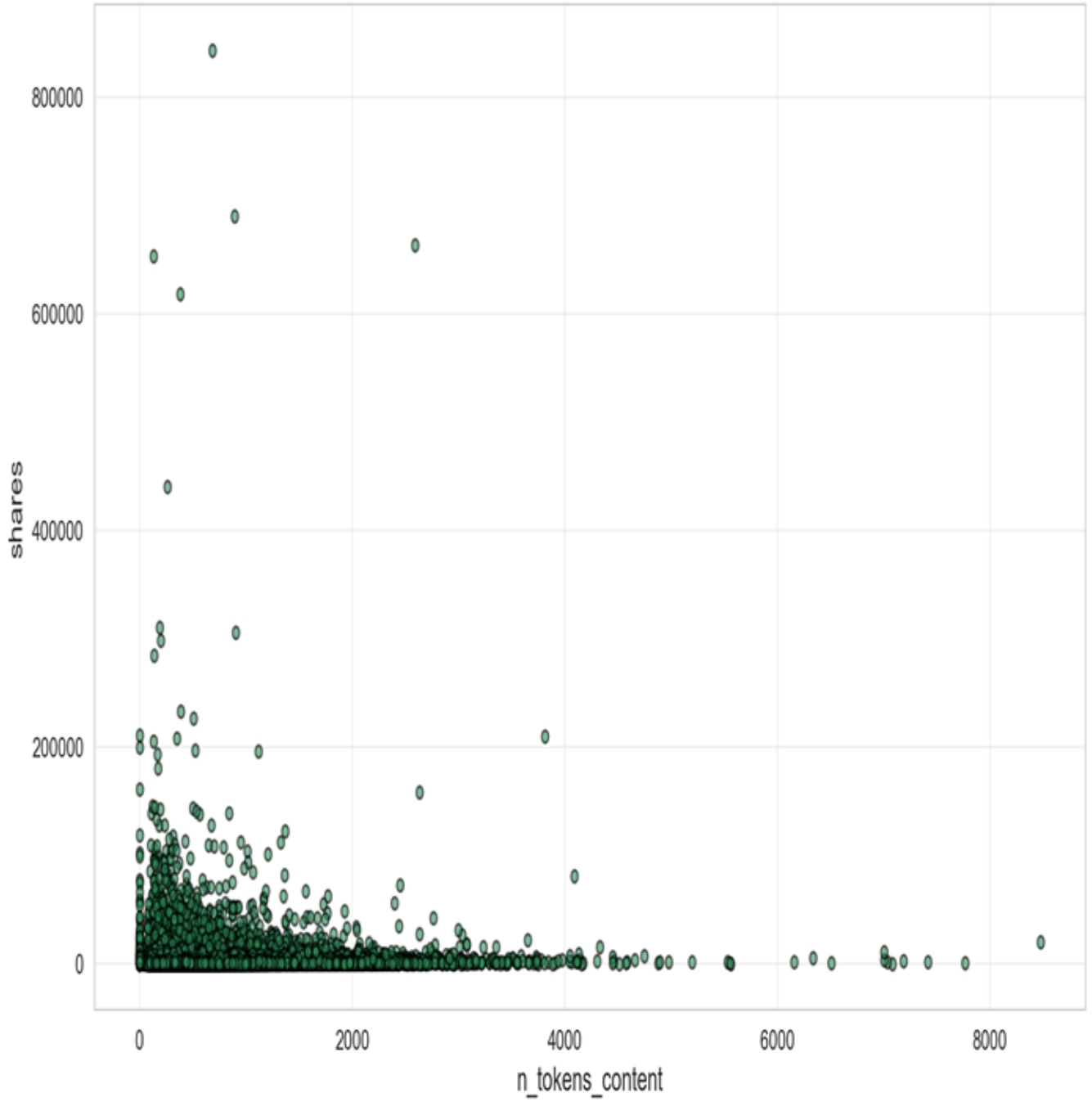


Figure 4: Scatter Plot — n_tokens_content vs shares



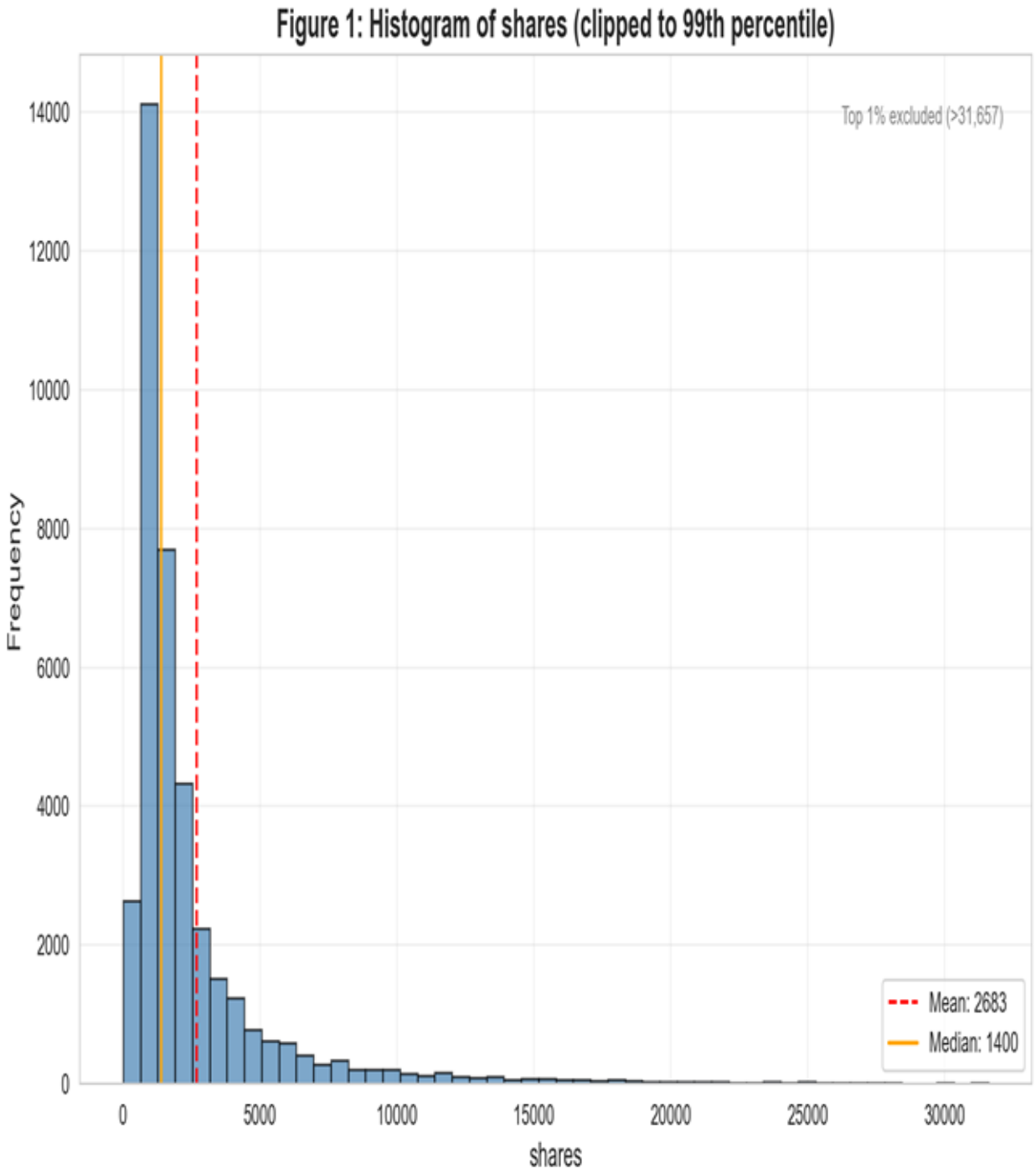


Figure 1: Histogram of shares (clipped to 99th percentile)
Top 1% excluded (>31,657)
Mean: 2683
Median: 1400
Frequency
shares
0
2000
4000
6000
8000
10000
12000
14000
0
5000
10000
15000
20000
25000
30000

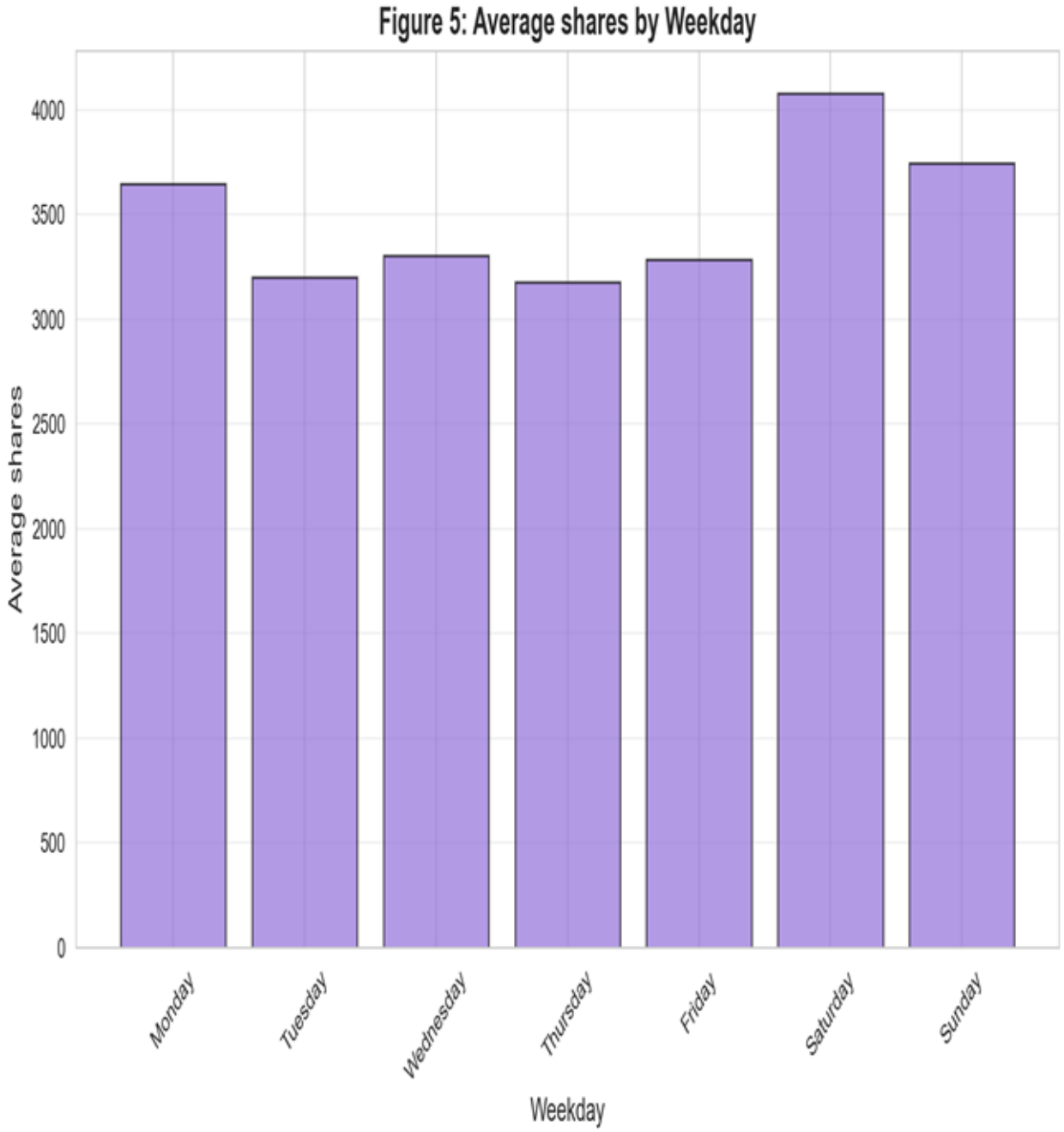


Figure 5: Average shares by Weekday
Average shares
Weekday
0
500
1000
1500
2000
2500
3000
3500
4000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

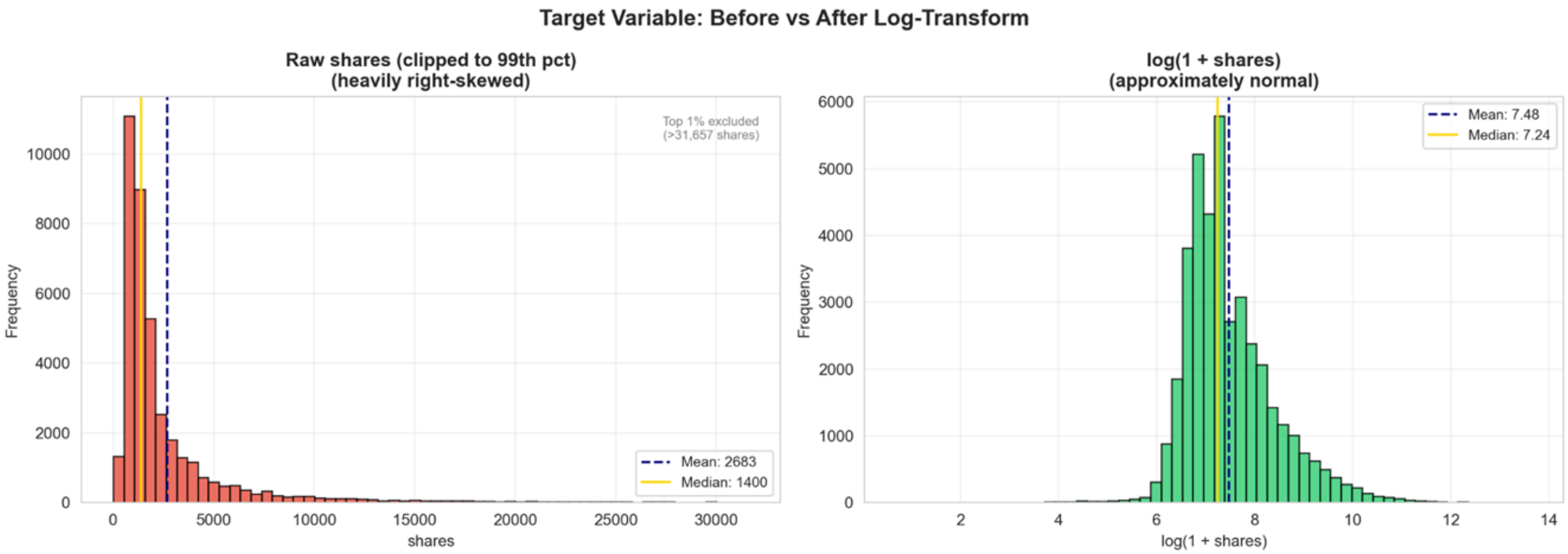
DATA PREPROCESSING STAGES

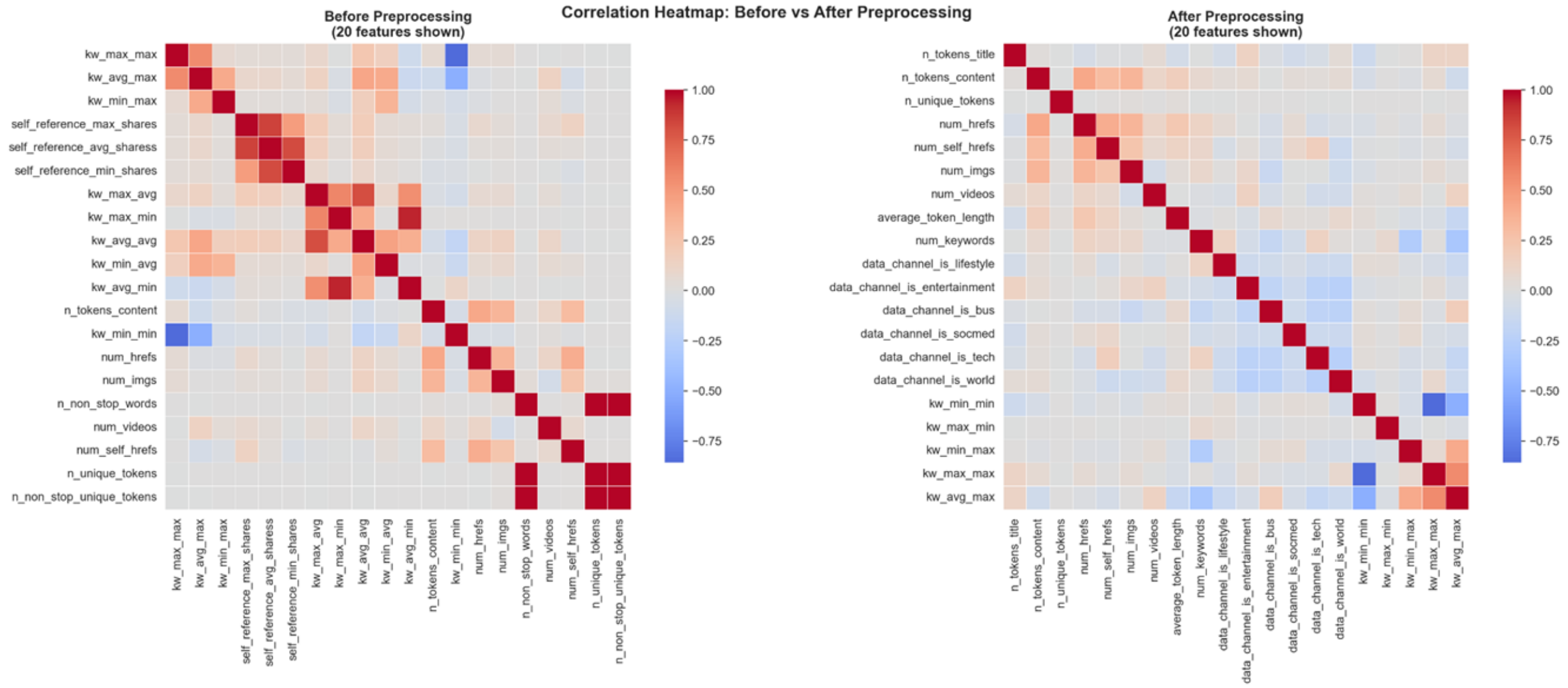
Basic Preprocessing

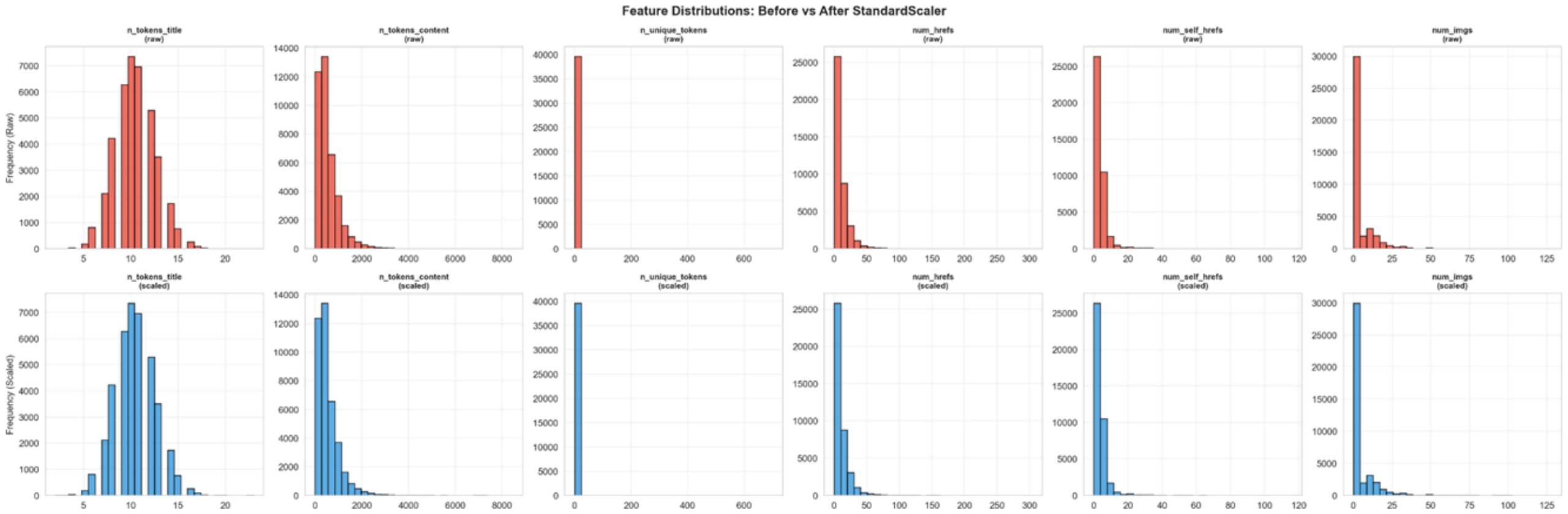
- Removed non-informative columns (URL)
- Correlation filter: Removed $|r| > 0.9$
- StandardScaler implementation

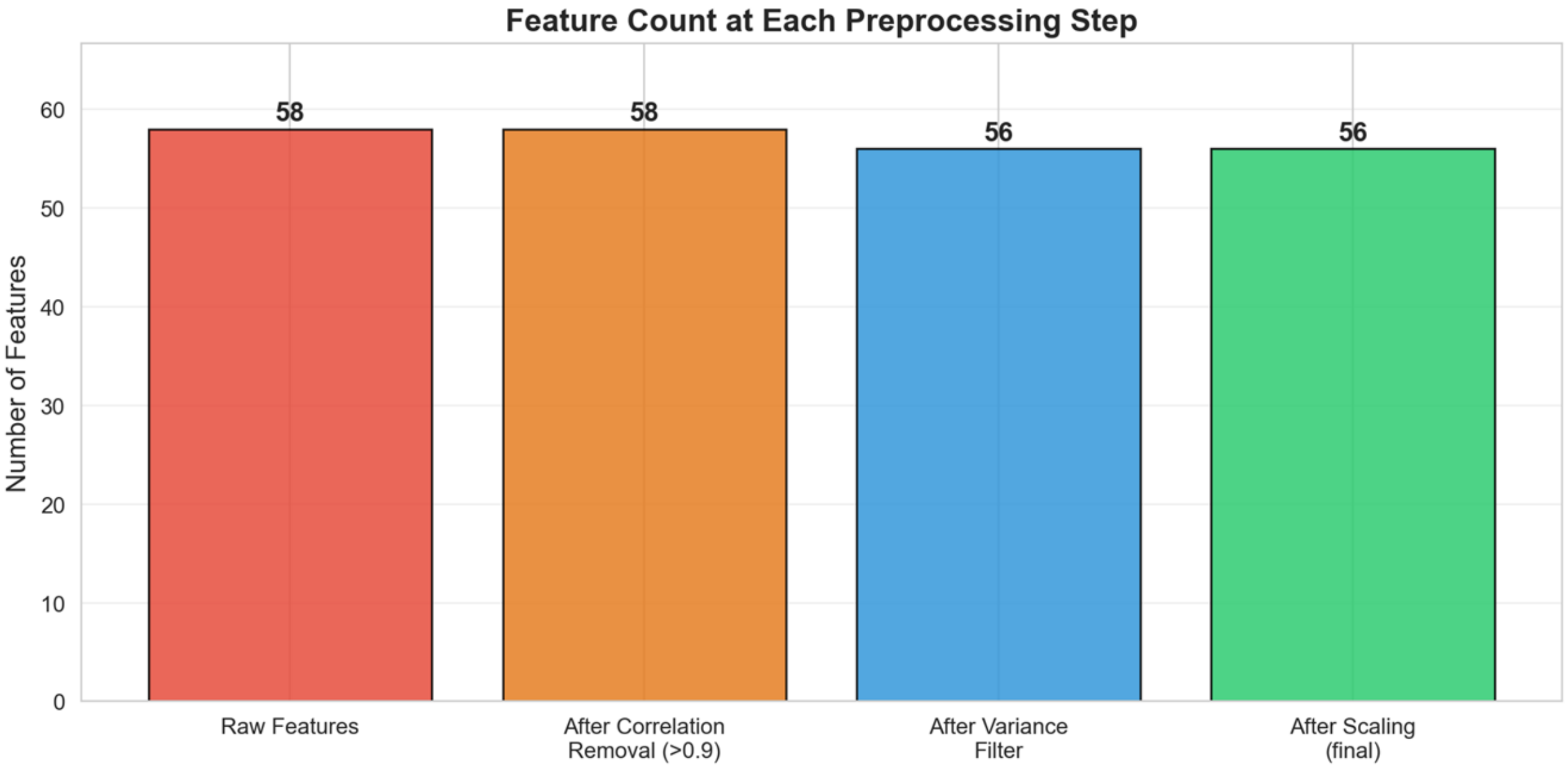
Advanced Pipeline

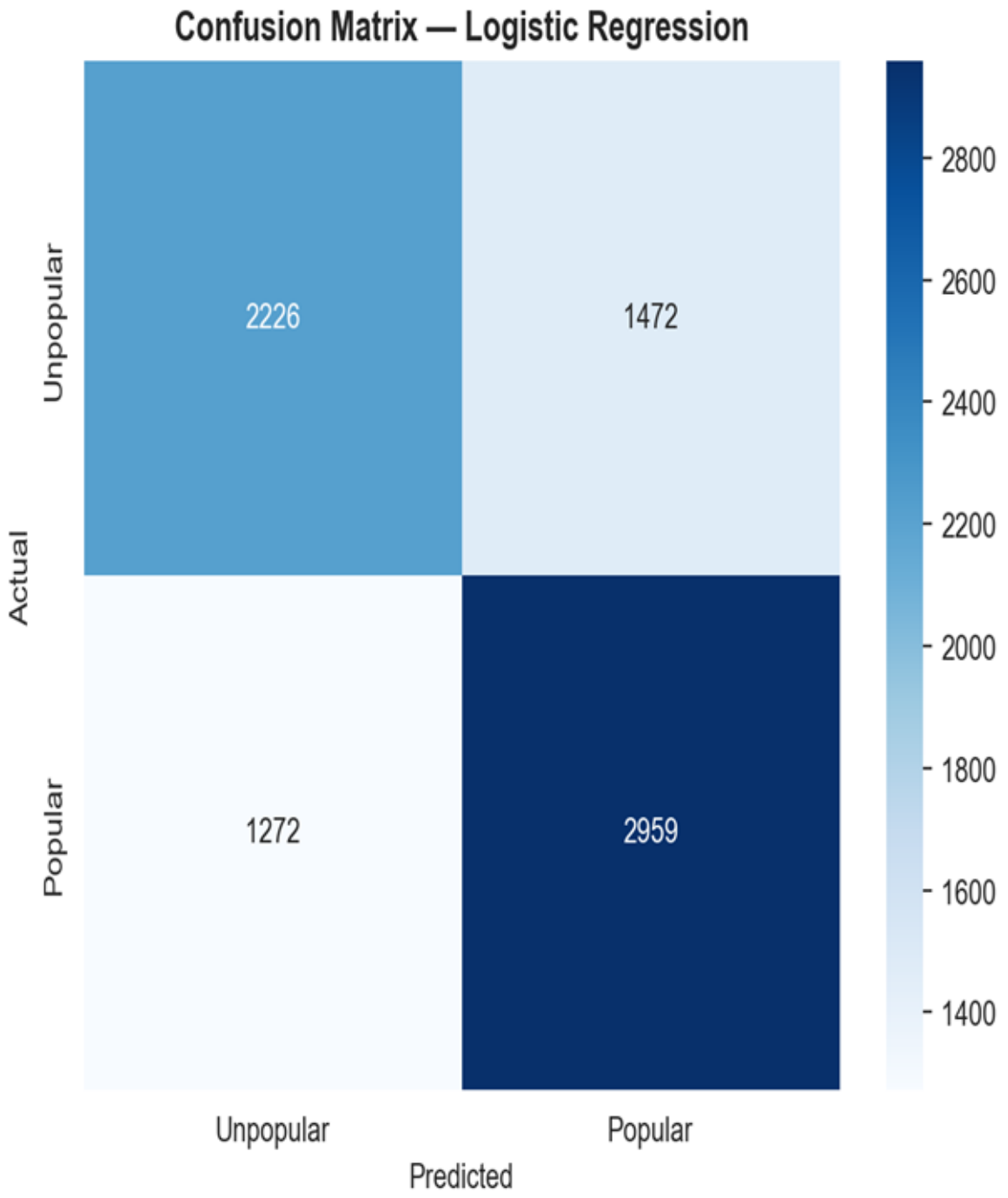
- Median Imputation for missing values
- Low-variance feature removal
- Correlation filtering and pruning
- 80/20 Train-Test split ratio

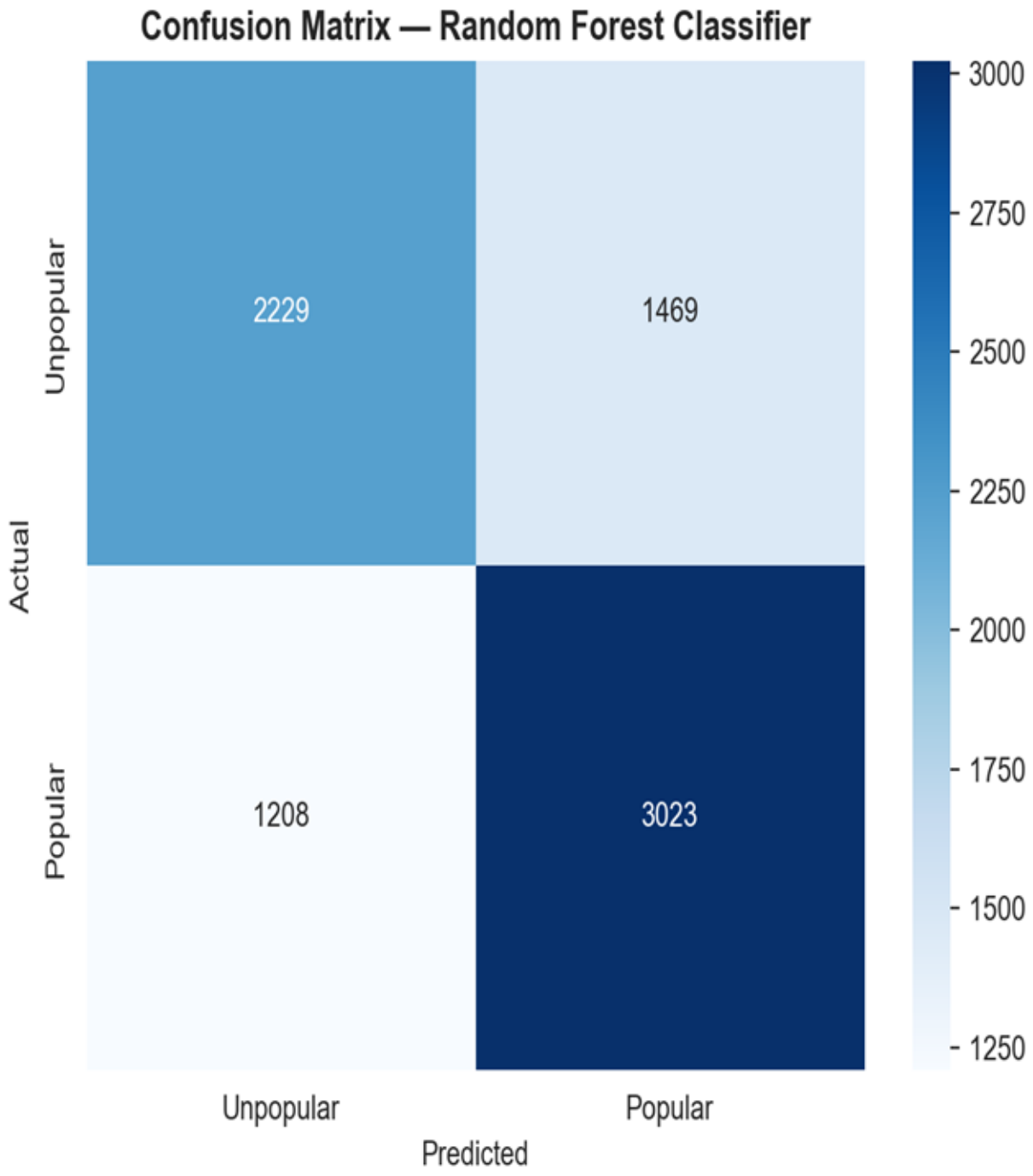

Target Variable: Before vs After Log-Transform
Raw shares (clipped to 99th pct)
(heavily right-skewed)
Top 1% excluded
(>31,657 shares)
Mean: 2683
Median: 1400
Frequency
shares
0
2000
4000
6000
8000
10000
0
5000
10000
15000
20000
25000
30000
log(1 + shares)
(approximately normal)
Mean: 7.48
Median: 7.24
Frequency
log(1 + shares)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14

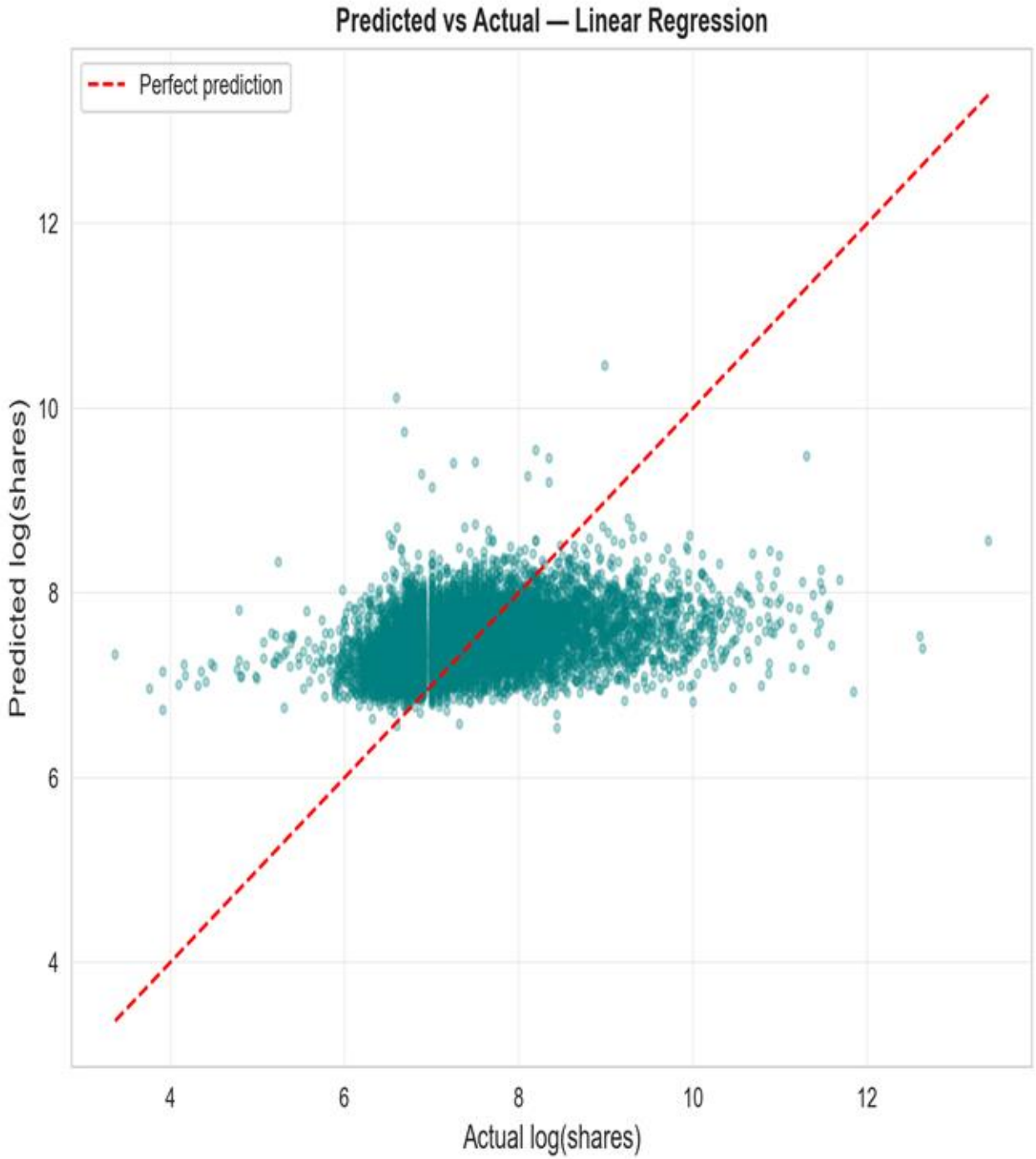

Correlation Heatmap: Before vs After Preprocessing
Before Preprocessing
(20 features shown)
kw_max_max
kw_avg_max
kw_min_max
self_reference_max_shares
self_reference_avg_sharess
self_reference_min_shares
kw_max_avg
kw_max_min
kw_avg_avg
kw_min_avg
kw_avg_min
n_tokens_content
kw_min_min
num_hrefs
num_imgs
n_non_stop_words
num_videos
num_self_hrefs
n_unique_tokens
n_non_stop_unique_tokens
After Preprocessing
(20 features shown)
n_tokens_title
n_tokens_content
n_unique_tokens
num_hrefs
num_self_hrefs
num_imgs
num_videos
average_token_length
num_keywords
data_channel_is_lifestyle
data_channel_is_entertainment
data_channel_is_bus
data_channel_is_socmed
data_channel_is_tech
data_channel_is_world
kw_min_min
kw_max_min
kw_min_max
kw_max_max
kw_avg_max
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

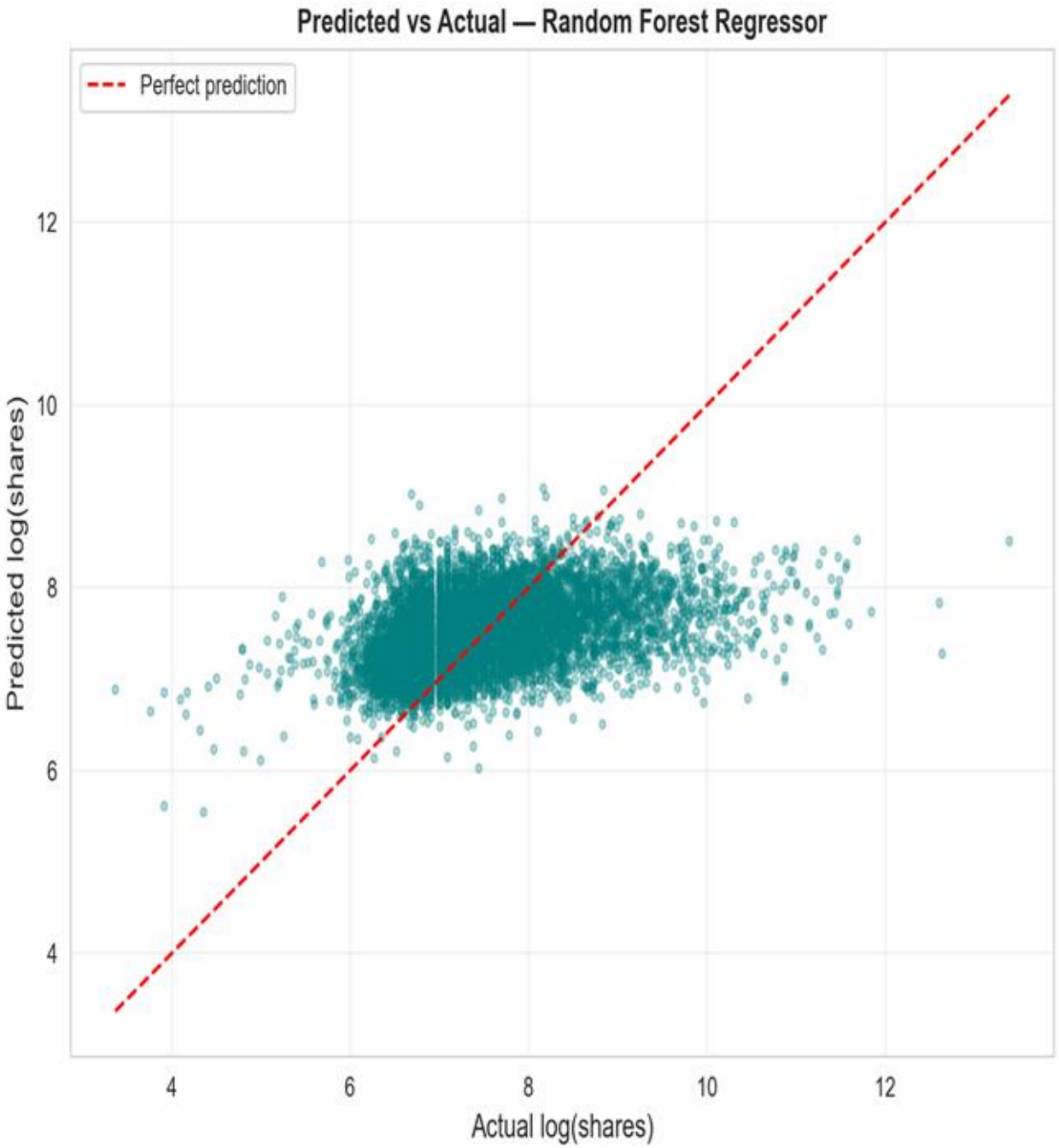

Feature Distributions: Before vs After StandardScaler
n_tokens_title (raw)
n_tokens_content (raw)
n_unique_tokens (raw)
num_hrefs (raw)
num_self_hrefs (raw)
num_imgs (raw)
n_tokens_title (scaled)
n_tokens_content (scaled)
n_unique_tokens (scaled)
num_hrefs (scaled)
num_self_hrefs (scaled)
num_imgs (scaled)
Frequency (Raw)
Frequency (Scaled)


Feature Count at Each Preprocessing Step
Number of Features
0
10
20
30
40
50
60
58
58
56
56
Raw Features
After Correlation Removal (>0.9)
After Variance Filter
After Scaling (final)


Confusion Matrix — Logistic Regression
Actual
Unpopular
Popular
2226
1472
1272
2959
Unpopular
Popular
Predicted
2800
2600
2400
2200
2000
1800
1600
1400


Confusion Matrix — Random Forest Classifier
Actual
Unpopular
Popular
2229
1469
1208
3023
Unpopular
Popular
Predicted
3000
2750
2500
2250
2000
1750
1500
1250


Predicted vs Actual — Linear Regression
Perfect prediction
Predicted log(shares)
Actual log(shares)
4
6
8
10
12


Predicted vs Actual — Random Forest Regressor
Perfect prediction
Predicted log(shares)
Actual log(shares)
4
6
8
10
12

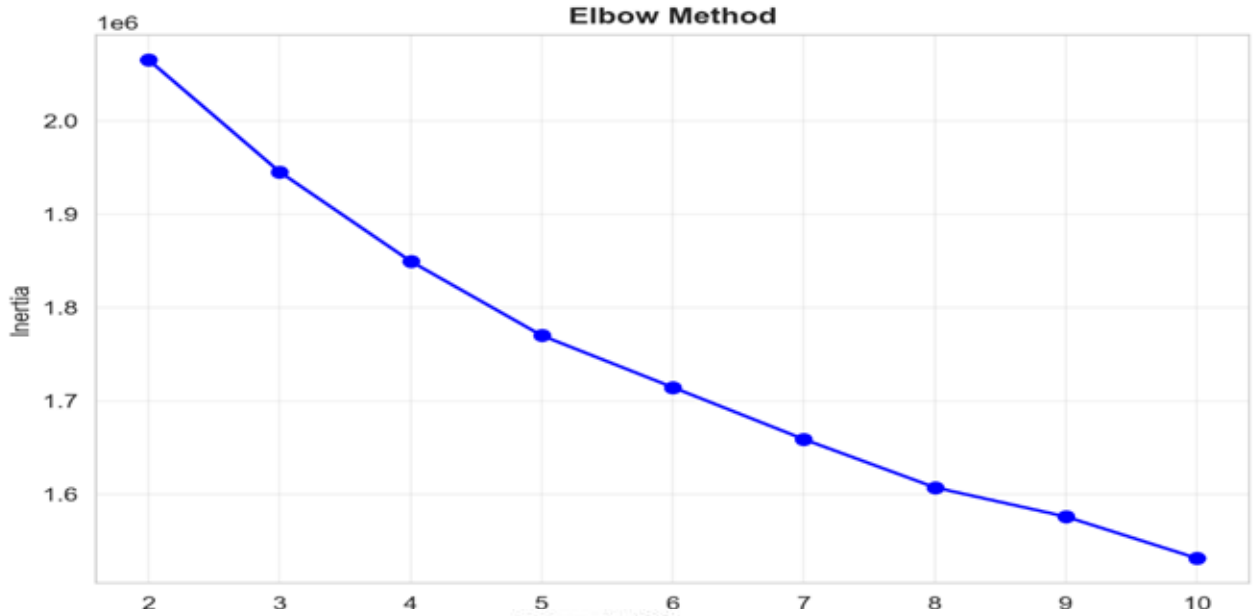
UNSUPERVISED CLUSTERING

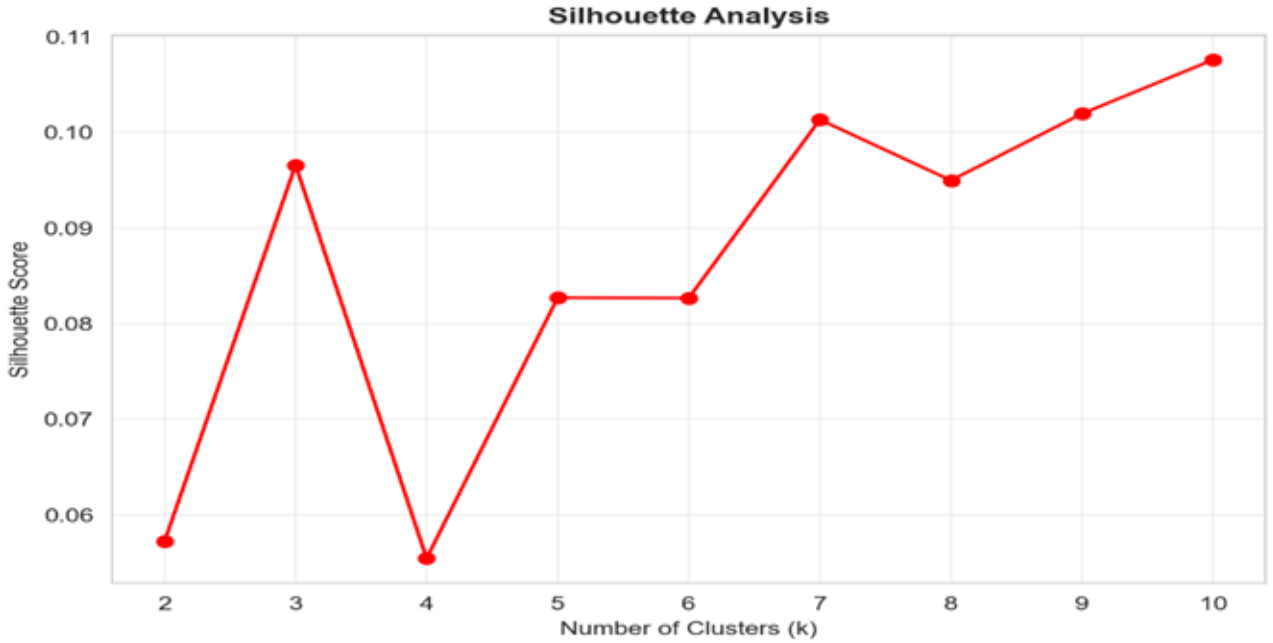
K-Means

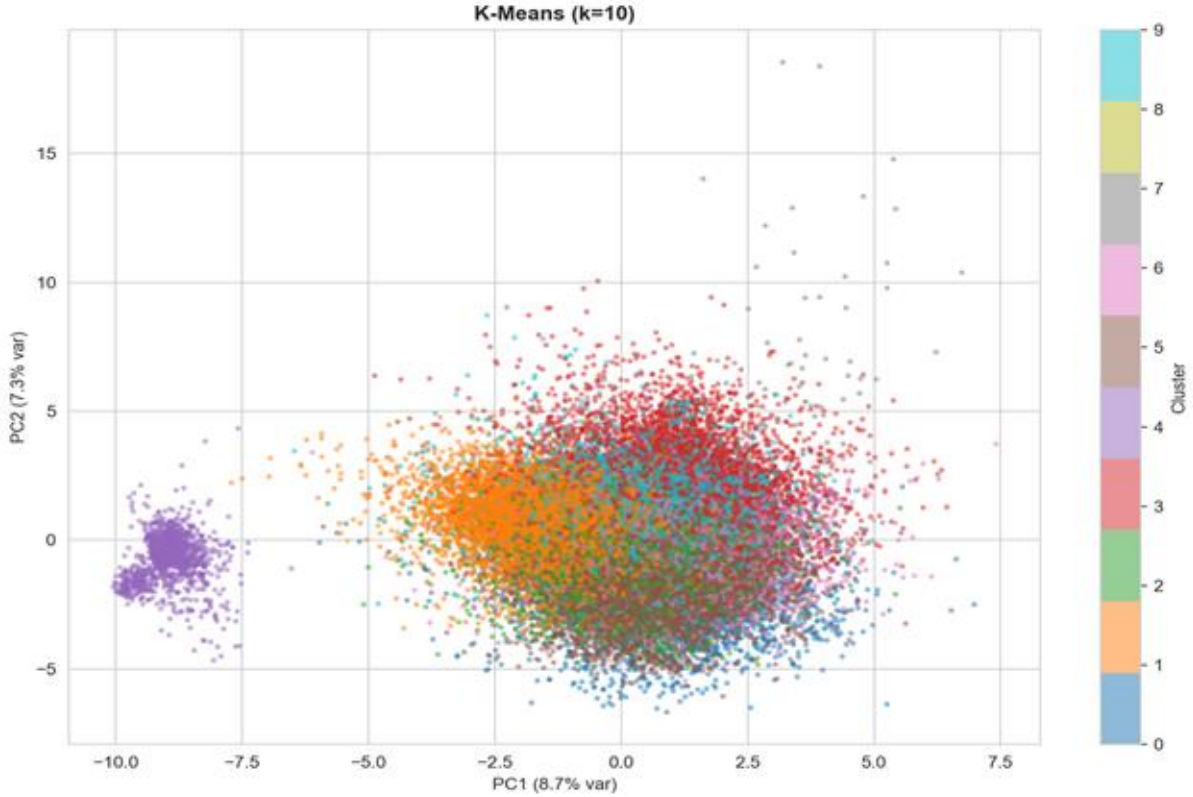
Groups articles into specific clusters based on feature similarity. Optimal 'K' determined via the **Elbow Method** and Silhoutte Score

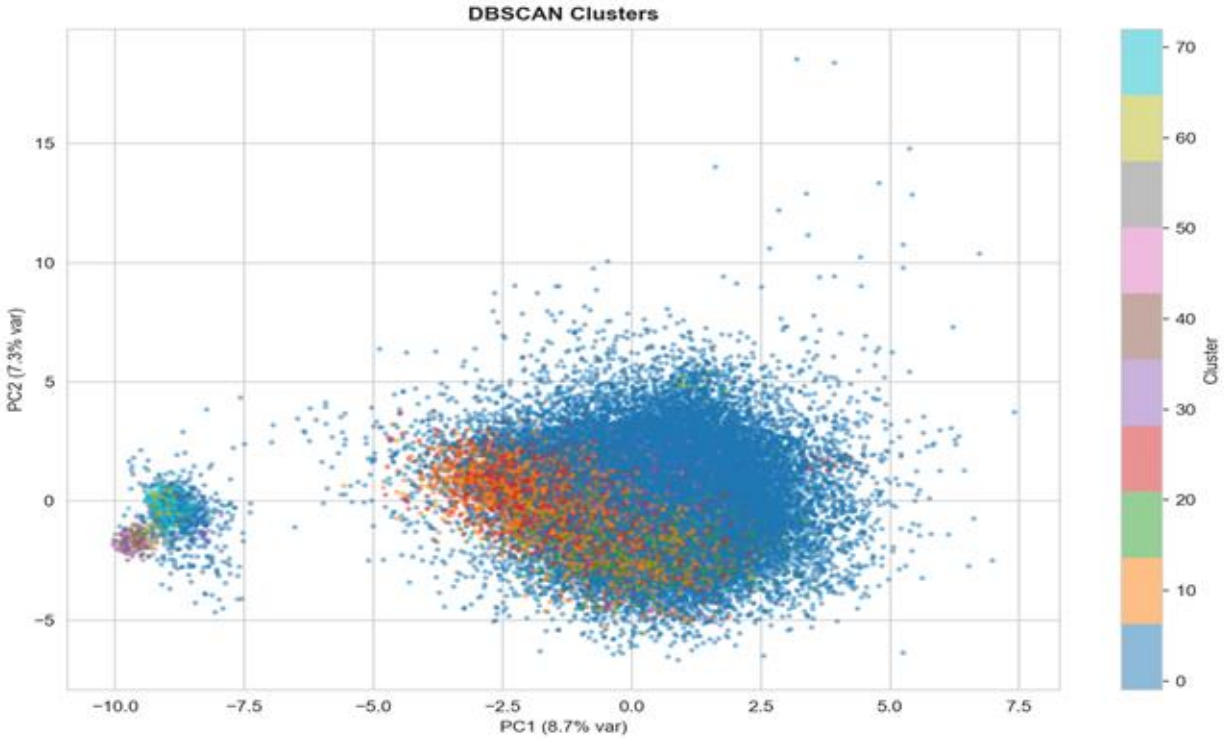
DBSCAN

Density-based clustering used to detect **Noise and Outliers** effectively in high-dimensional space.


Elbow Method
1e6
Inertia
2.0
1.9
1.8
1.7
1.6
2
3
4
5
6
7
8
9
10


Silhouette Analysis
Silhouette Score
0.11
0.10
0.09
0.08
0.07
0.06
2
3
4
5
6
7
8
9
10
Number of Clusters (k)


K-Means (k=10)
PC2 (7.3% var)
15
10
5
0
−5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
PC1 (8.7% var)
Cluster
9
8
7
6
5
4
3
2
1
0


DBSCAN Clusters
PC2 (7.3% var)
15
10
5
0
−5
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
PC1 (8.7% var)
Cluster
70
60
50
40
30
20
10
0

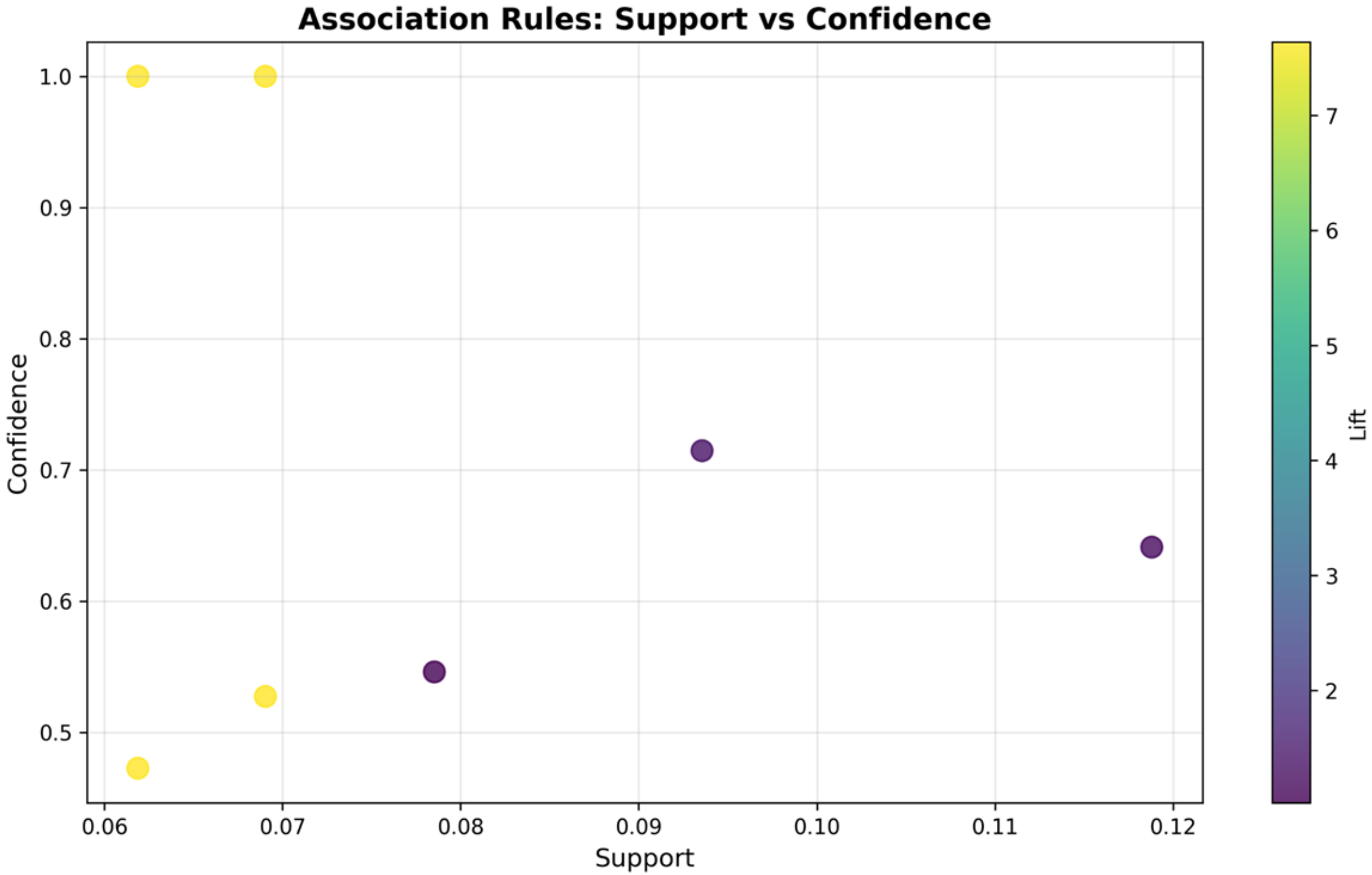
ASSOCIATION RULE MINING

Apriori

Mining Metrics

- **Support:** Rule frequency
- **Confidence:** Reliability
- **Lift:** Significance of relationship

Purpose: Identify specific feature combinations (e.g., weekend + specific topic) associated with higher social engagement.


Association Rules: Support vs Confidence
Confidence
1.0
0.9
0.8
0.7
0.6
0.5
0.06
0.07
0.08
0.09
0.10
0.11
0.12
Support
Lift
7
6
5
4
3
2

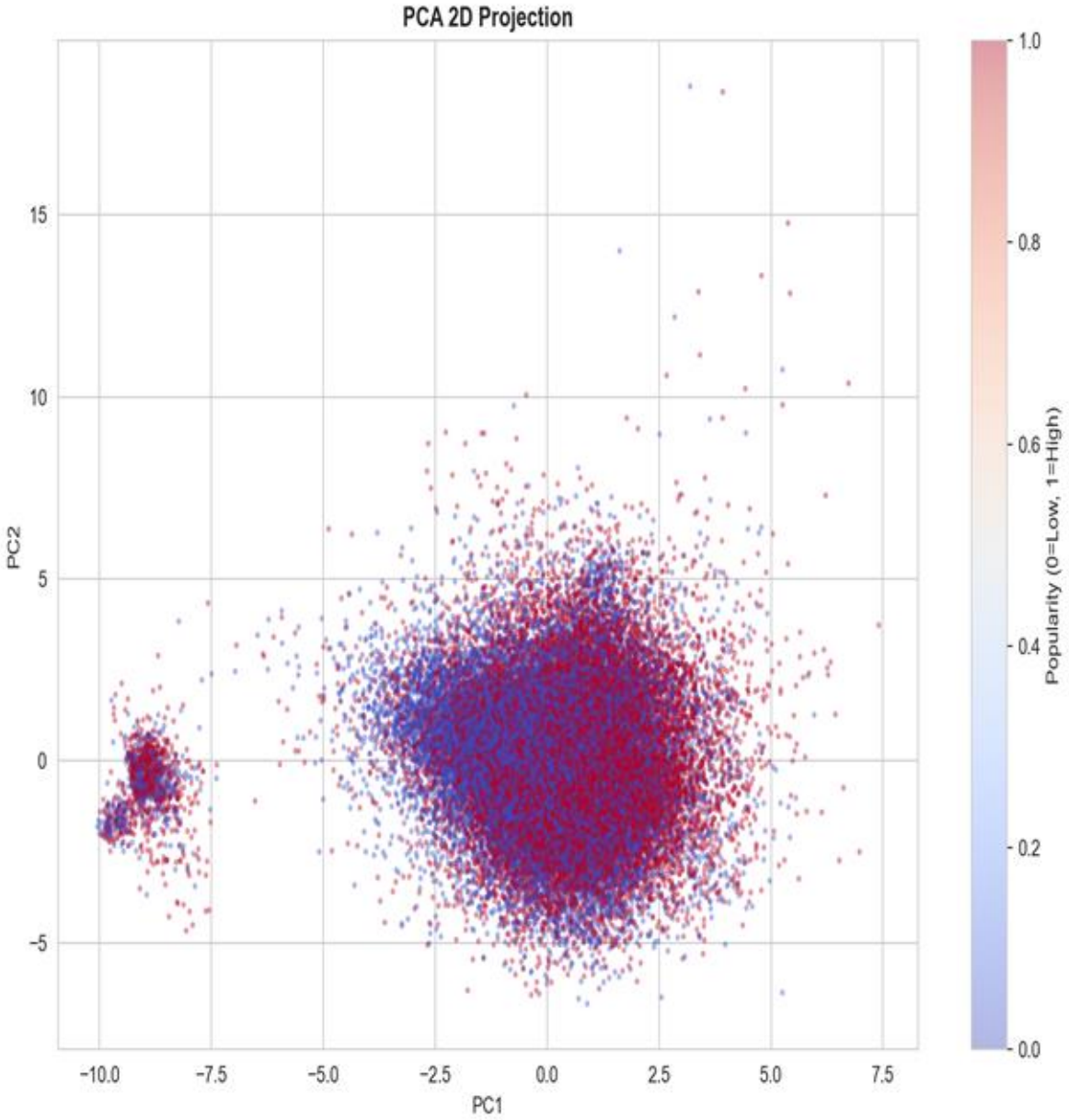
DIMENSIONALITY REDUCTION

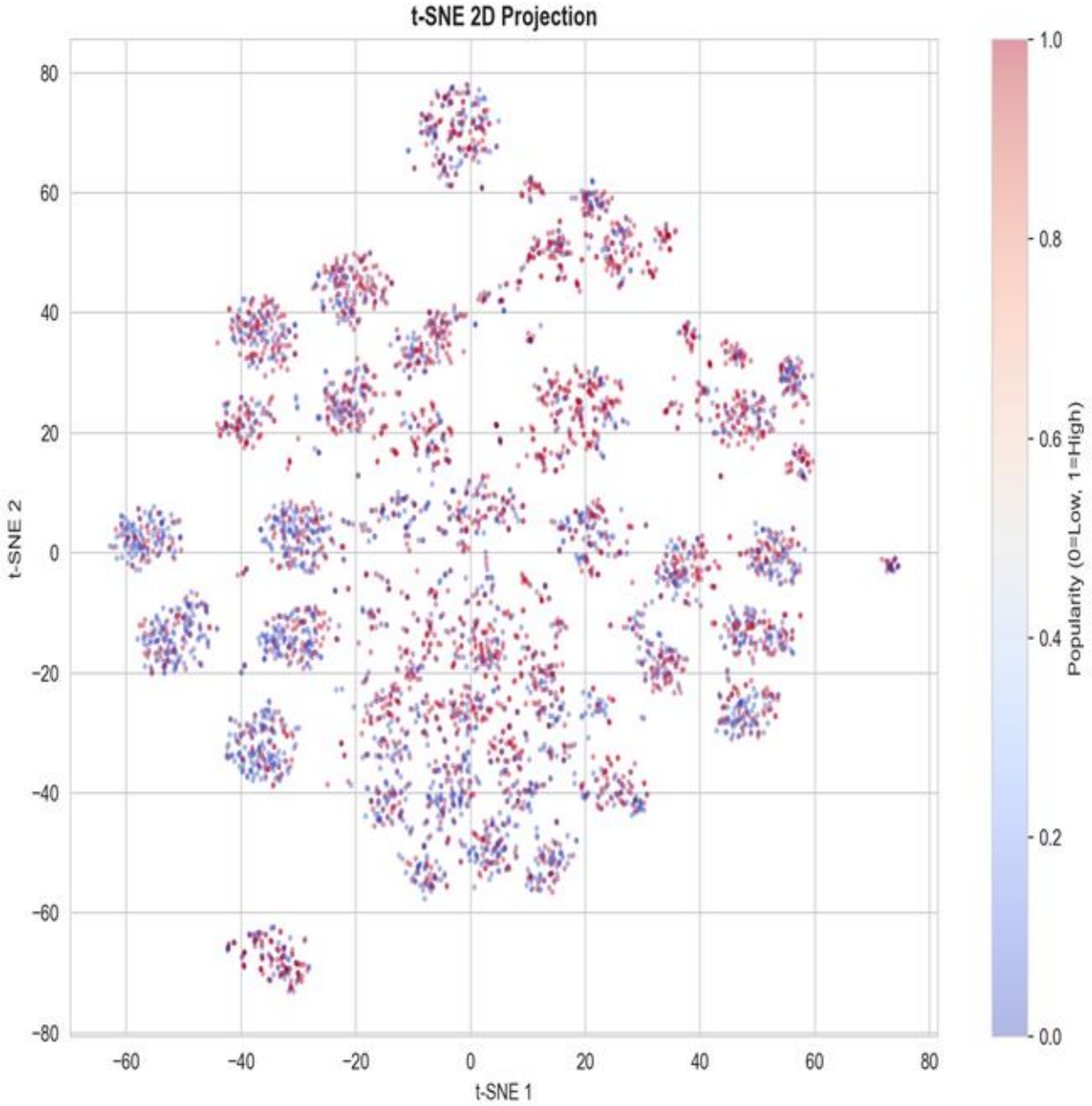
PCA

Principal Component Analysis was utilized to reduce dimensions while preserving the maximum variance in the data, facilitating computation.

t-SNE

t-distributed Stochastic Neighbor Embedding was used for visualizing local patterns and manifold structures in high-dimensional feature space.


PCA 2D Projection
PC1
PC2
Popularity (0=Low, 1=High)

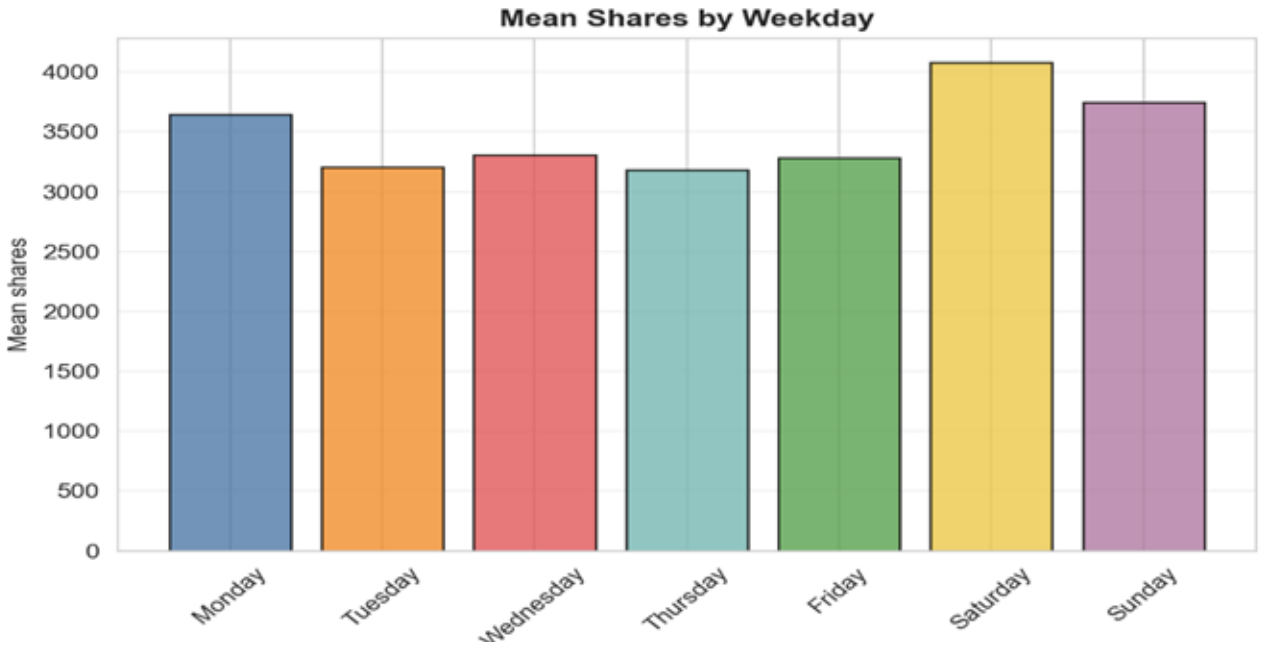

t-SNE 2D Projection
t-SNE 1
t-SNE 2
Popularity (0=Low, 1=High)

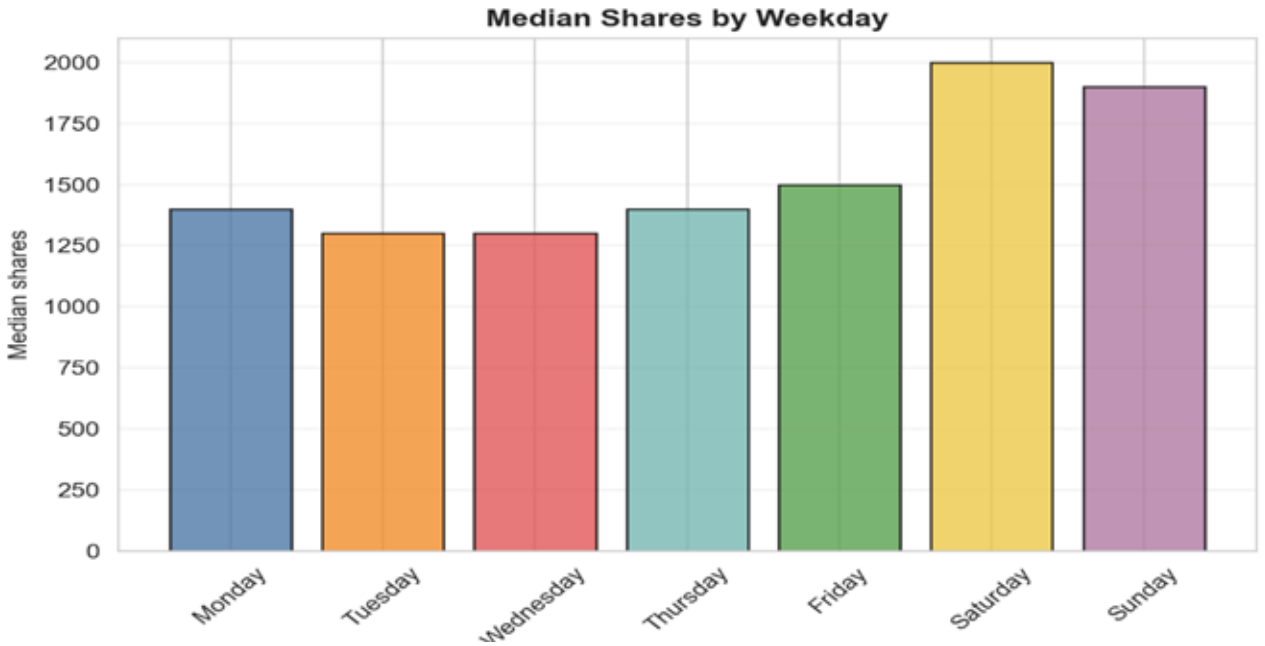
Temporal Analysis

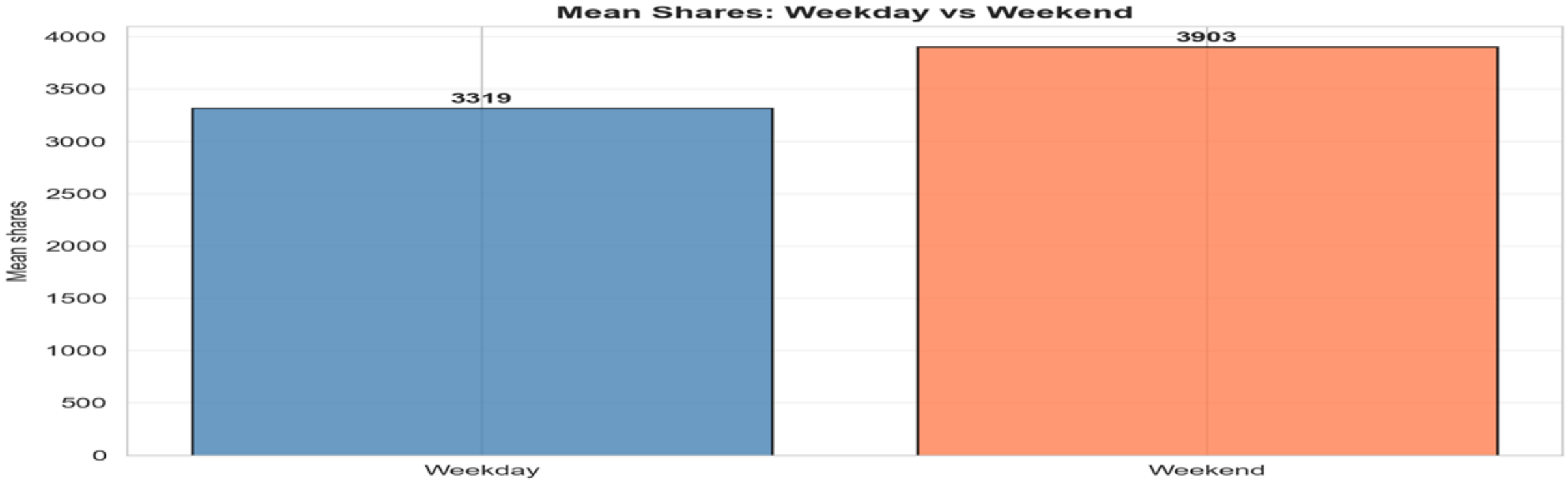
- Analyze how **day of publication** affects article shares

Approach

- Compared **average & median shares** across weekdays
- Grouped data into **weekday vs weekend**

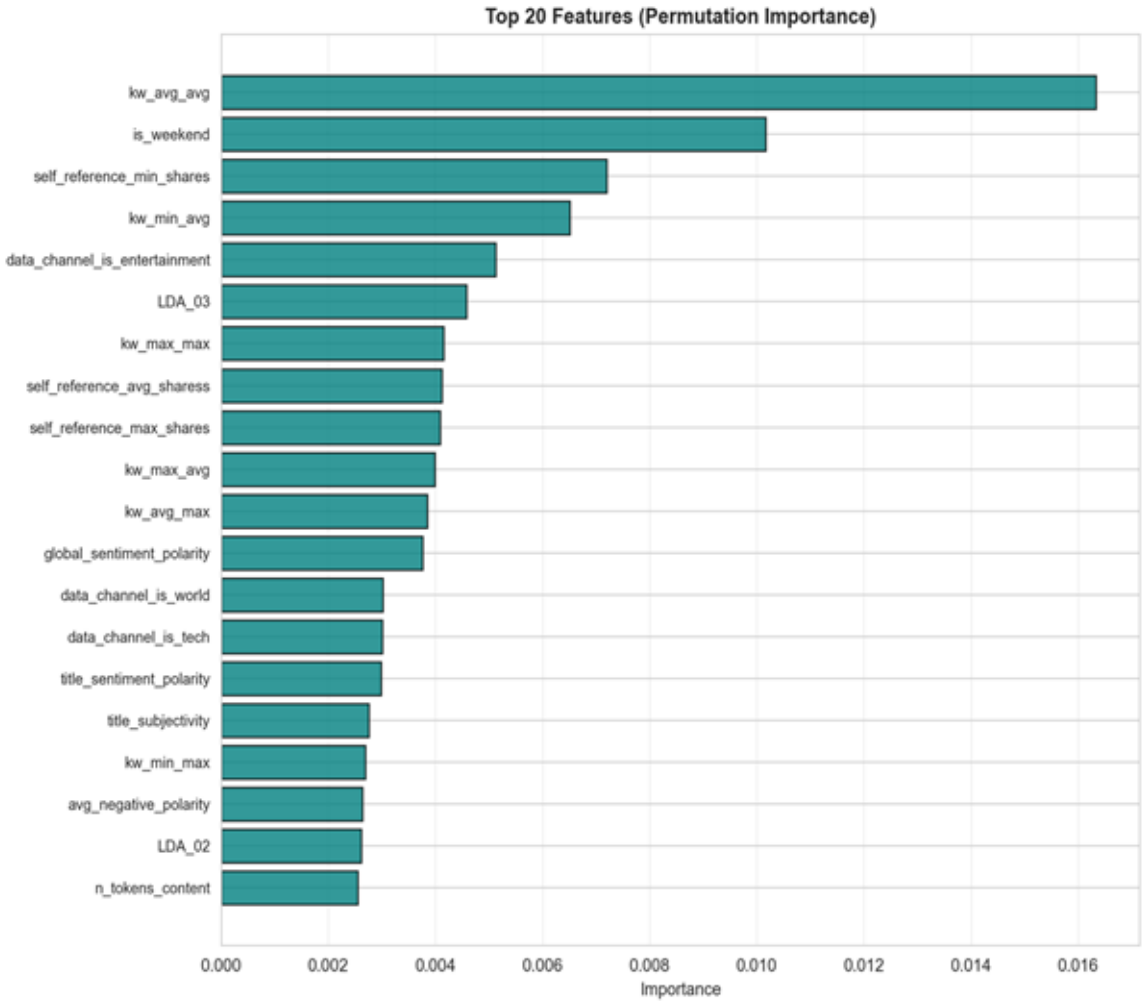

Mean Shares by Weekday
Mean shares
0
500
1000
1500
2000
2500
3000
3500
4000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday

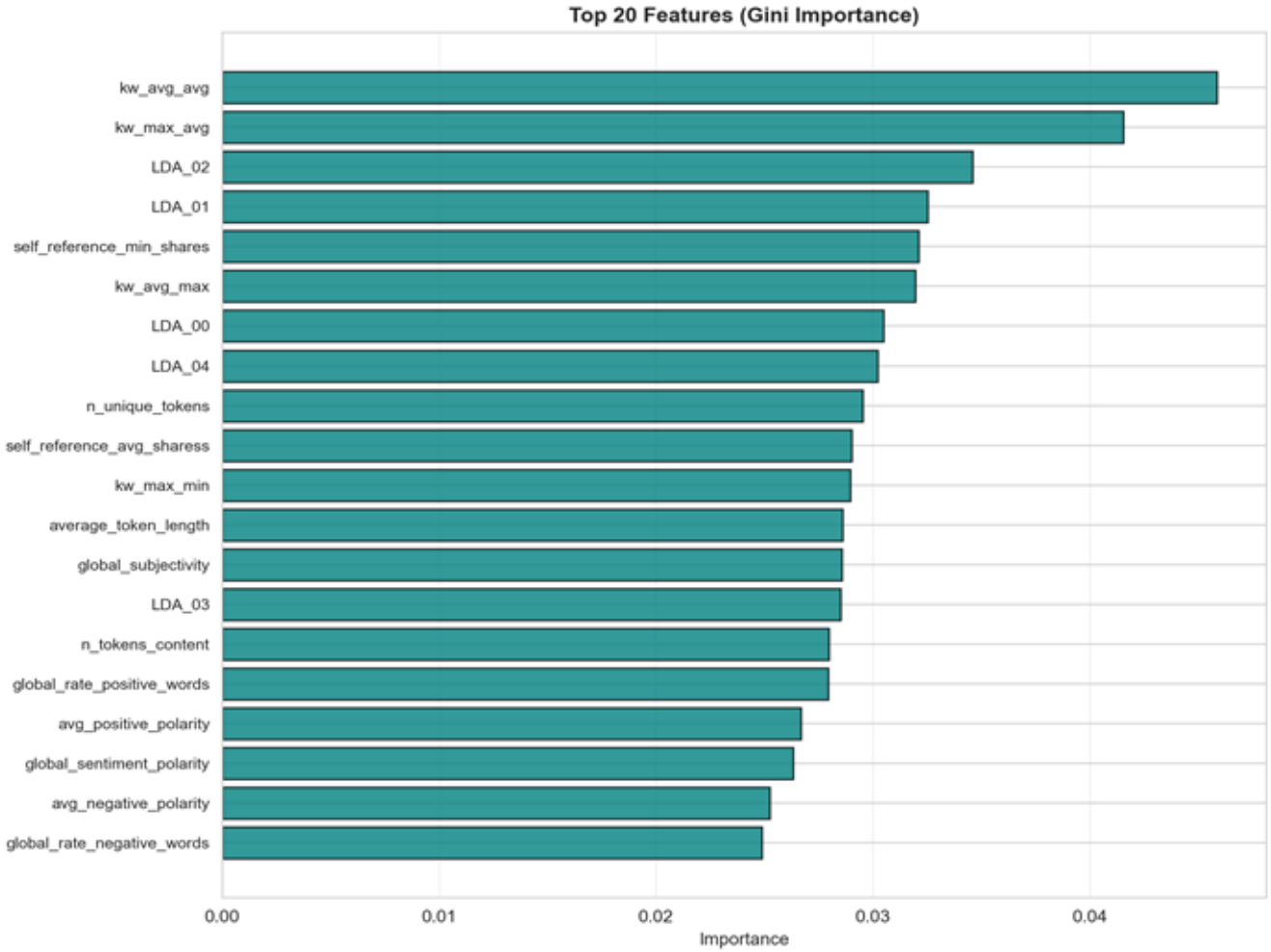

Median Shares by Weekday
Median shares
0
250
500
750
1000
1250
1500
1750
2000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Mean Shares: Weekday vs Weekend
Mean shares
0
500
1000
1500
2000
2500
3000
3500
4000
3319
3903
Weekday
Weekend

Feature Importance

Measures how much each feature contributes to model predictions . Helps identify **most influential variables** affecting popularity





CRITICAL PREDICTORS

Evaluated using **Random Forest Gini Importance** and **Permutation Importance** methods.

Controlled Preprocessing Experiment

All classification and regression models were trained under two conditions:

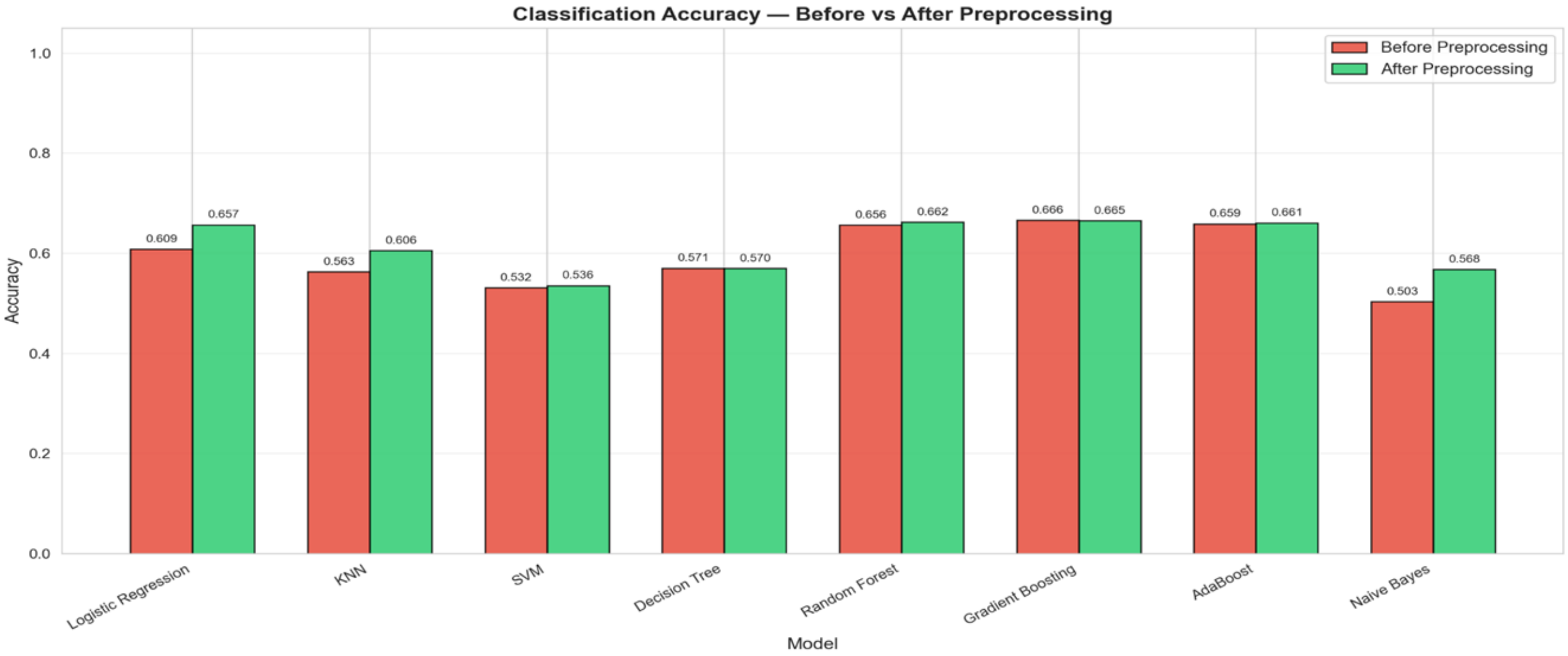
1. Using the raw feature set
2. Using the fully preprocessed feature set

Classification (8 models)

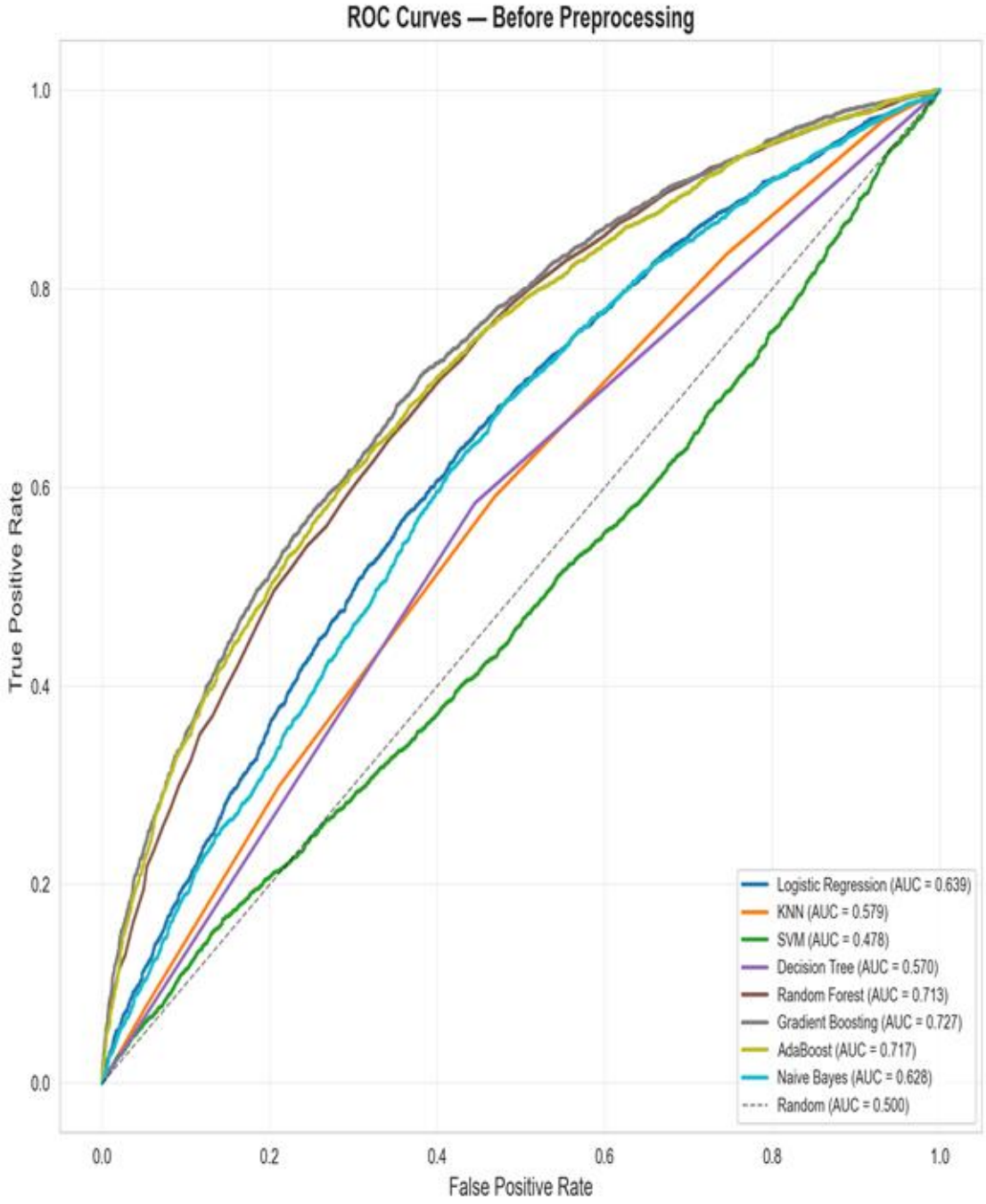
- LR, KNN, SVM, RF, Decision Tree, Gradient boosting, Adaboost, Naive Bayes

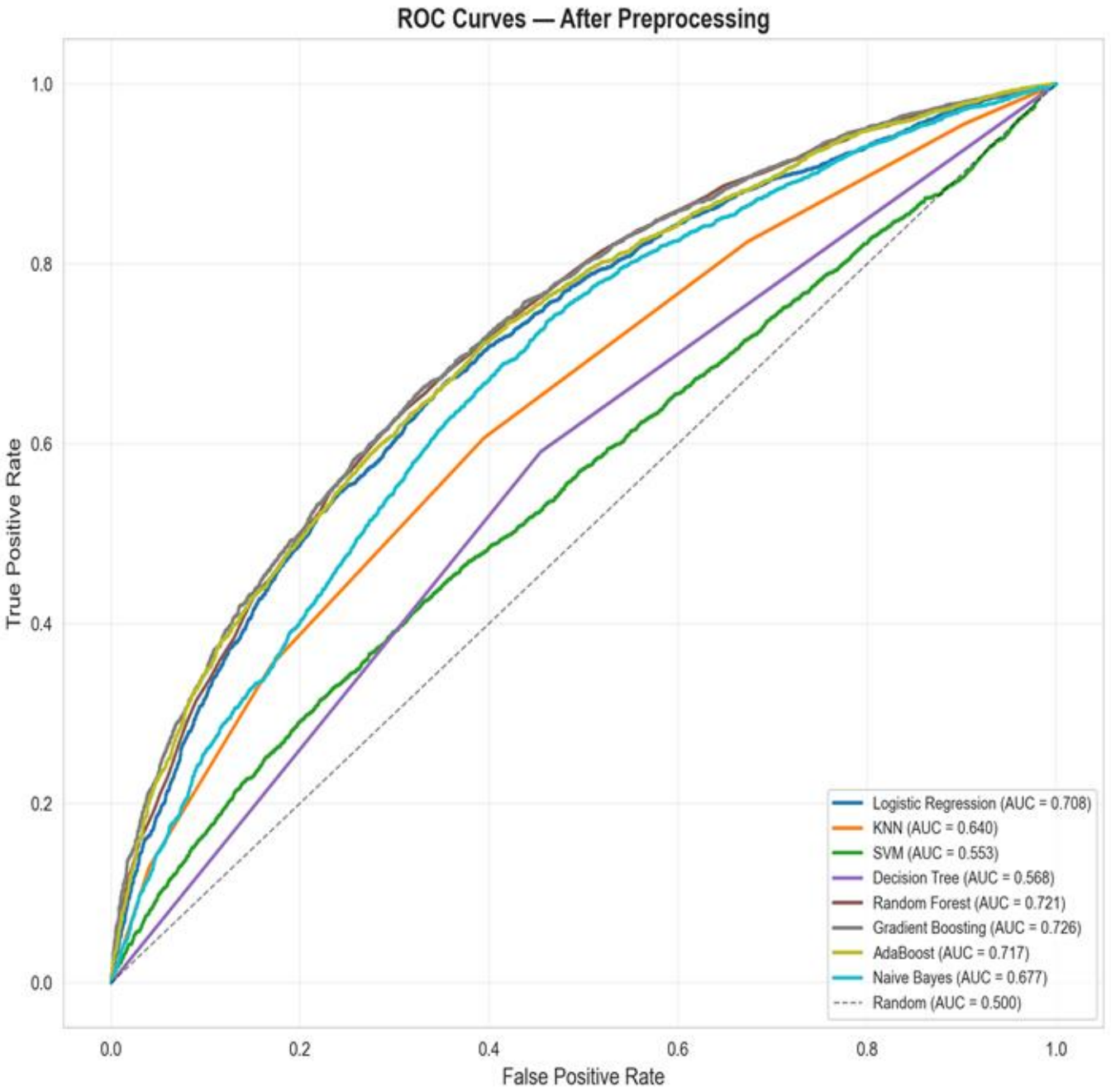
Regression (7 models)

- Linear Regression, Ridge Regression, Lasso Regression, Decision Tree Regressor, Gradient Boosting Regressor, SVR



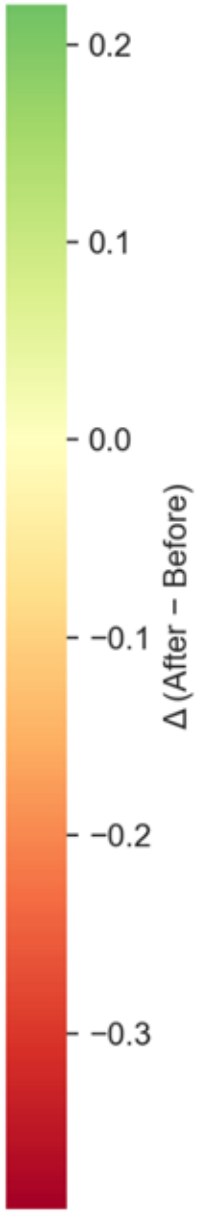
Evaluation Metrics : Accuracy, Precision/Recall, F1 Score, ROC-AUC


ROC Curves — Before Preprocessing
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.639)
KNN (AUC = 0.579)
SVM (AUC = 0.478)
Decision Tree (AUC = 0.570)
Random Forest (AUC = 0.713)
Gradient Boosting (AUC = 0.727)
AdaBoost (AUC = 0.717)
Naive Bayes (AUC = 0.628)
Random (AUC = 0.500)


ROC Curves — After Preprocessing
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.708)
KNN (AUC = 0.640)
SVM (AUC = 0.553)
Decision Tree (AUC = 0.568)
Random Forest (AUC = 0.721)
Gradient Boosting (AUC = 0.726)
AdaBoost (AUC = 0.717)
Naive Bayes (AUC = 0.677)
Random (AUC = 0.500)

Classification — Metric Change After Preprocessing (Δ)

	Accuracy	Precision	Recall	F1	ROC_AUC
Logistic Regression	0.0485	0.0530	0.0030	0.0302	0.0693
KNN	0.0429	0.0469	0.0156	0.0308	0.0616
SVM	0.0040	0.0278	-0.3888	-0.1130	0.0743
Decision Tree	-0.0009	-0.0024	0.0069	0.0023	-0.0014
Random Forest	0.0058	0.0070	-0.0002	0.0036	0.0087
Gradient Boosting	-0.0011	-0.0006	-0.0021	-0.0013	-0.0012
AdaBoost	0.0018	0.0014	0.0024	0.0018	0.0000
Naive Bayes	0.0649	0.0333	0.1832	0.2202	0.0491



REGRESSION MODELING

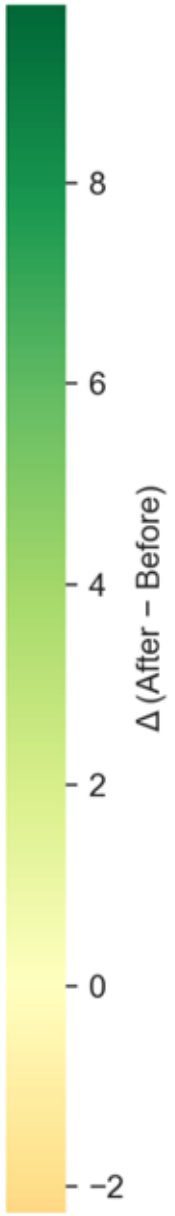
Target Transformation: To handle skewness, the target variable was transformed.

$$\text{Target} = \log(1 + \text{shares})$$

Model Type	Specific Algorithms	Metrics
Linear Models	Linear, Ridge, Lasso Regression	RMSE
Tree Models	Decision Tree, Random Forest	MAE
Advanced	Gradient Boosting, SVR	R^2 Score

Regression — Metric Change After Preprocessing (Δ)

	RMSE	MAE	R2
Linear Regression	-0.0012	-0.0004	0.0025
Ridge Regression	-0.0012	-0.0002	0.0024
Lasso Regression	-0.0047	-0.0044	0.0094
Decision Tree Regressor	0.0045	-0.0007	-0.0127
Random Forest Regressor	0.0015	0.0017	-0.0030
Gradient Boosting Regressor	0.0004	0.0002	-0.0008
SVR	-2.1657	-2.2586	9.7777



FINAL MODEL RESULTS

Task	Best Model	Key Metric Value
Classification	Random Forest	Accuracy: 0.6624
Regression	Random Forest Regressor	RMSE: 0.8493

Results indicate that **nonlinear relationships** significantly influence news popularity, favoring ensemble tree-based models over linear alternatives.

PREPROCESSING IMPACT

- **Significant Gains:** Logistic Regression, KNN, and Naive Bayes saw major performance boosts.
- **Stability:** Tree-based models (RF, Boosting) showed minimal changes as they are robust to feature scaling.
- **Standardization:** Critical for SVR convergence and metric accuracy.

Preprocessing is not universally beneficial — its impact depends on the type of model used.

SCALABLE PYSPARK PIPELINE

Load

Spark DataFrames ingestion of 40k records.

Assemble

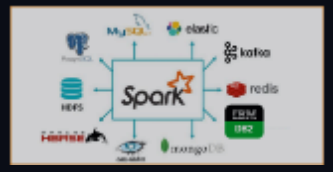
VectorAssembler for feature consolidation.

Scale

StandardScaler within Spark ML pipeline.

Train

Distributed RandomForest training.



CHALLENGES & FUTURE WORK

Key Challenges

- High Dimensionality (60 features)
- Extreme target skewness
- Multicollinearity issues
- Extensive parameter tuning

Future Directions

- Improve **feature engineering** for better insights
- Multi-class virality tier prediction
- Data Augmentation for pyspark

CRITICAL ANALYTICAL INSIGHTS

Non-linear Superiority

Ensemble models (RF, Boosting) outperform linear ones, indicating complex relationships in feature interactions.

Predictive Difficulty

Low R^2 values (~0.15) suggest that popularity is highly volatile and influenced by unpredictable external factors.

Skewness Impact

Severe distribution skewness (virality vs. average) required **Log Transformation** for model stability.

Strongest Predictors

Keyword statistics and LDA topic features show the highest importance across all supervised learning models.

Timing Factors

Weekend publication consistently yields higher engagement, reflecting distinct social sharing behavior patterns.

Robustness

Tree-based models are more robust. Less sensitive to preprocessing and scaling. Suitable for real-world noisy data

ANALYTICAL INSIGHTS II

Weak Clustering

Low silhouette scores show articles do not form clear groupings; popularity is not cluster-driven.

Preprocessing Variance

Distance models (KNN, LR) benefit greatly from preprocessing and scaling, while Tree-based models remain insensitive.

Behavioral Patterns

Association rules confirm specific "Tech + Weekend" combinations maximize share probability.

Separability Issues

PCA/t-SNE reveal overlapping data points, making classification boundaries extremely hard to define.

Outliers influence

Viral articles act as extreme outliers

Can distort model learning, especially regression

Prediction Trade-off

Classification performs better than regression, indicating it is easier to predict popularity trends than exact share counts.

Conclusion

- Popularity is multifactorial (Content + Topic + Timing).
- Ensemble models are the most effective predictors.
- Preprocessing is mandatory for linear/distance models (model-dependent)
- Publication timing and content features influence engagement.
- Predicting exact shares is difficult due to data complexity
- Big Data tools (Spark) ensure scalability.

DEMO

Questions?

Thank you for your attention.